Berkeley APR - Research Proposal - March

Links:
Berkeley APR - Possibility Lab
Berkeley APR - HCD - definitions
Berkeley APR - Claude - Workflow - March
City of Berkeley Housing Element APR Annual Progress Report

Berkeley APR - Session Summary - March 23

Research Collaboration Proposal: Open Housing Data Infrastructure

From: John Gage, Berkeley Housing Pipeline Project

Dear Professor Lerman, Professor O'Neill, and colleagues,

I'm writing to propose a research collaboration on improving the quality, accessibility, and civic utility of California housing production data — specifically the Annual Progress Report (APR) process administered by California State Housing and Community Development department.

What We've Built

Berkeley Housing Pipeline Explorer

Datasette open source data journalism database

Live at: https://berkeley-housing.fly.dev/

The API is fully functional with faceted search, canned queries, and map visualization. Researchers can now query the data directly via:

- Web interface: https://berkeley-housing.fly.dev/berkeley_housing_map
- JSON API: https://berkeley-housing.fly.dev/berkeley_housing_map/projects.json
- SQL queries: https://berkeley-housing.fly.dev/berkeley_housing_map?sql=...

Over the past several months, I've developed an independent, open-source data pipeline that collects, validates, and analyzes Berkeley's housing permit data. The project now tracks 162 active housing projects representing over 10,000 units, drawn directly from primary sources (the Accela permitting system, Alameda County parcel records, and GIS spatial data). The pipeline produces APR-format tables that can be compared project-by-project against the city's official submissions to HCD.

Key components:

- A structured data collection workflow using Berkeley's Accela Citizen Access portal, with parsers that extract permit timeline events, fee payments, and document inventories

- A SQLite database with 29,024 parcels, 65,459 addresses, and 42 zoning districts spatially joined to development potential analysis
- An APR specification validator (based on HCD's field definitions for Tables A, A2, B, and C)
- A series of 40 Jupyter notebooks organized as a teachable data science curriculum, deployed as Google Colab notebooks for classroom use
- Comparison tooling that matches our independent dataset against HCD's published APR data from the California Open Data Portal

What We Found

Our independent audit of Berkeley's 2024 APR revealed several categories of discrepancy:

Missing projects in both directions. Our dataset contains projects that the city omitted from its APR submission, and the city's report contains projects we had not yet collected. Neither dataset is independently complete. A merged, cross-validated dataset would be more accurate than either source alone.

Systematic data collection gaps. Berkeley's Accela system contains all the raw data needed for APR reporting, but there is no automated pipeline from Accela to HCD's APR format. City staff manually compile the report, introducing the same kinds of errors Professor O'Neill documented in San Francisco — incorrect year assignments, missing entitlements, and inconsistent unit counts.

Income data opacity. Affordable unit counts (VLI/LI/MOD) by deed restriction status are buried in PDF attachments (Density Bonus Eligibility Statements) rather than stored as structured data in the permitting system. Our automated extraction from permit descriptions captured only 18% of the affordable units the city reported. This suggests the city itself relies on manual document review for this critical RHNA tracking data. Our data now includes URL links to every PDF document filed by any applicant with the city, and all City reviews and revisions, allowing us to ingest every public document held by the city.

Construction start and project stalling are invisible. We identified 19 approved or pending projects representing 1,464 units with no permit activity in over 12 months. There is no systematic way to distinguish "approved and proceeding" from "approved and stalled" using public data alone. This gap matters for accurate RHNA progress forecasting. We now can actively detect stalling projects, providing a rich resource for evaluating the actual effectiveness of specific ordinances and State laws.

The Opportunity: A Civic Data Quality Infrastructure

The Possibility Lab's partnership with HCD to improve APR data quality, combined with the Terner Center's rigorous methodology for validating APR submissions, creates an opportunity to build something more systemic than one-off audits.

We propose exploring a collaborative approach with several components:

1. Student-Powered Data Validation

Our Jupyter notebook curriculum is designed for deployment in high school and community college data science classes. Students working through the notebooks perform authentic civic research: collecting permit data, cross-referencing it against official reports, calculating processing timelines, and identifying discrepancies. Each student who audits 10 permits contributes to a collective dataset that improves APR accuracy.

This model scales: every city in California uses similar permitting systems (Accela, Clariti/OpenGov, TRAKiT). The notebook framework can be adapted to any jurisdiction by changing the data source configuration. A statewide network of student data analysts could systematically validate APR submissions across hundreds of jurisdictions.

2. Open Data Infrastructure Requirements

We advocate for a policy framework that would make independent APR validation routine rather than heroic:

Municipal open data ordinances for housing data. Berkeley and other cities should be required to publish building permit data through their open data portals (Socrata, ArcGIS Hub, or equivalent) in machine-readable formats with daily or weekly updates. Berkeley currently publishes business licenses and parcel data but has no building permit dataset available for programmatic access.

Vendor requirements for permitting software. Accela, Clariti/OpenGov, and other permitting platforms should be contractually required to provide API access to permit records, status histories, and document metadata. The data exists in these systems — it simply isn't exposed. Procurement specifications for permitting software should include open API requirements as a condition of purchase.

Standardized affordability data fields. Income unit counts (VLI/LI/MOD), deed restriction status, and density bonus calculations should be stored as structured data fields in the permitting system, not buried in PDF attachments. This single change would make APR Table A2 generation largely automatable.

Datasette as a lightweight publication platform. We use Datasette (an open-source tool for publishing SQLite databases as interactive, queryable websites) to make our housing data browsable and API-accessible. Cities could adopt Datasette as a low-cost complement to their existing data portals, providing instant API access to housing pipeline data without procurement cycles.

3. Performance Measurement

Our timeline data enables metrics that aren't currently tracked in the APR:

- **Filing-to-complete duration**: How long does the completeness review take? (Our data shows ranges from 1 month to 13+ months for similar project types)
- **Complete-to-entitled duration**: How long from deemed complete to approval?
- **Entitlement-to-building-permit duration**: How long between approval and construction authorization?
- **Stalled project identification**: Which approved projects show no activity?
- **Developer performance**: Which development teams move projects through the pipeline efficiently?
- **Planner workload distribution**: Which staff members are assigned to the most projects?

These metrics serve multiple constituencies: city managers evaluating process efficiency, housing advocates tracking RHNA progress, developers planning project timelines, and researchers studying regulatory barriers to housing production.

What We're Asking

We would like to continue our conversations about how our open-source toolkit, your research expertise, and HCD's institutional partnership could combine to create a scalable approach to housing data quality. Specifically:

1. **Methodology review**: Would the Terner Center or Possibility Lab be interested in reviewing our independent Berkeley dataset against the city's APR submission, similar to Professor O'Neill's San Francisco analysis?
2. **Student deployment pilot**: Could we collaborate on deploying the notebook curriculum in a UC Berkeley undergraduate or graduate housing data course, with students auditing Berkeley's (or another jurisdiction's) APR data as a course project? The existing 40 Jupyter notebook curriculum is designed for adoption by California high schools in every municipality. Perhaps linking these data science initiatives could provide a million California high school students with a collaborative vision of UC Berkeley data science.
3. **Policy framework development**: Would the Possibility Lab's HCD partnership be an appropriate channel for advancing open data requirements for permitting systems in every California city?
4. **Statewide scaling**: How might this approach extend beyond Berkeley to other jurisdictions in the ABAG region or statewide?

Resources

- Berkeley Housing Pipeline data and notebooks:
  - https://blockxblock.github.io/berkeley-housing-analysis/explorer.html
  - https://berkeley2050.github.io/guide/
- YouTube channel with project walkthroughs: @BuildBerkeley2050
- APR specification validator and gap analysis tools: available on GitHub
- Datasette deployment of Berkeley housing data: deployed on Fly.io

  - Live at: https://berkeley-housing.fly.dev/

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  - SQL queries: https://berkeley-housing.fly.dev/berkeley_housing_map?sql=...

I look forward to hearing your thoughts.

Best regards, John Gage Berkeley, California

📝 Notes

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- Berkeley APR - Possibility Lab
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- 2026-03-23-Monday
- Nico Savidge
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Notes last touched today

- 2026-03-22-Sunday
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